Ultra-Thin Flexible Printed Circuits

Ultra-Thin Flexible Printed Circuits

Ultra-Thin Flexible Printed Circuits









CONTACT AN ENGINEER

Ultra-Thin Flexible Printed Circuits (FPC) for Lightweight, Compact Electronics

Flexible Circuit Boards Engineered for Minimal Thickness, Maximum Performance

Ultra-thin flexible printed circuits let you push into ultra-compact form factors where every gram and micron count. PICA delivers precision FPCs with ultra-thin materials and advanced manufacturing for wearables, medical devices, foldables and space-constrained applications.

Learn more about how to overcome the challenges in designing ultra-thin flex circuits. Read more here.

Why Choose Ultra-Thin Flexible Printed Circuits?

- **Minimal Thickness & Weight** – Ultra-thin substrates allow board thicknesses down to ~0.20 mm or less, enabling slimmer, lighter assemblies.
- **Space-Constrained Packaging** – Ideal for wearable tech, medical implants, foldable electronics and any application where height is restricted or comfort is key.
- **High Flexibility & Conformability** – Ultra thin construction enables tight bend radii, fold-to-install, even wraparound designs.
- **Precision Manufacturing & Materials** – Achieved through adhesive-less polyimide stacks, laser-direct imaging, and specialized handling for ultra-thin panels.

- **Optimized for Miniaturization** – Supports fine-line routing, small pad designs, low mass interconnects for next-gen devices.

Ultra-Thin FPC Capabilities – Highlights

- Ultra-thin constructions: e.g., 3 µm copper on 25 µm polyimide; 18 µm copper on 12.5 µm polyimide; finished thickness as low as 0.20 mm ± 0.08 mm.
- Adhesiveless / thermoplastic polyimide (TPI) material sets for improved durability and precision.
- Laser Direct Imaging (LDI) and fine-line etching for ultra-thin substrate processing and trace precision.
- Specialized process controls for ultra-thin panel handling: thickness/flatness, scratch resistance, warpage mitigation.
- Design guidance: low mass, low profile circuits for wearables, sensors, compact modules, with engineering collaboration for manufacturability and reliability.

Global Design & Manufacturing Support

PICA's engineering team works closely with your design team—from concept to volume production—to define ultra-thin stack-ups, material selection, bend features, flatness tolerances and assembly readiness. Early collaboration ensures manufacturability, reliability and efficient production of ultra-thin flex circuits for your application.

Benefits of Ultra-Thin Flexible Circuits

Ultra-Low Profile & Lightweight
These circuits enable form-factors previously impossible—devices are thinner, lighter and more comfortable without sacrificing performance.

Markets We Serve with Ultra-Thin Flexible Circuits

Wearables & Consumer Devices
Smart watches, fitness bands, AR/VR headsets and ultra-portable devices rely on ultra-thin flex circuits to meet form-factor and comfort demands.

Enhanced Flexibility & Design Freedom
Ultra-thin construction supports tight bends, non-standard layouts and space-optimized routing, perfect for wearables, foldables and compact modules.

Miniaturization Without Compromise
By reducing the board thickness and weight, you free up space for more features or smaller enclosures —accelerating innovation in next-gen electronics.

Reduced Mass & Mechanical Stress
Less material means lower inertia during motion, less stress on solder joints and better overall durability in dynamic or wearable systems.

Streamlined Manufacturing & Supply Chain
With PICA's mature processes for ultra-thin flex manufacturing, you benefit from production-ready reliability, yield assurance and scalability from prototype to volume.

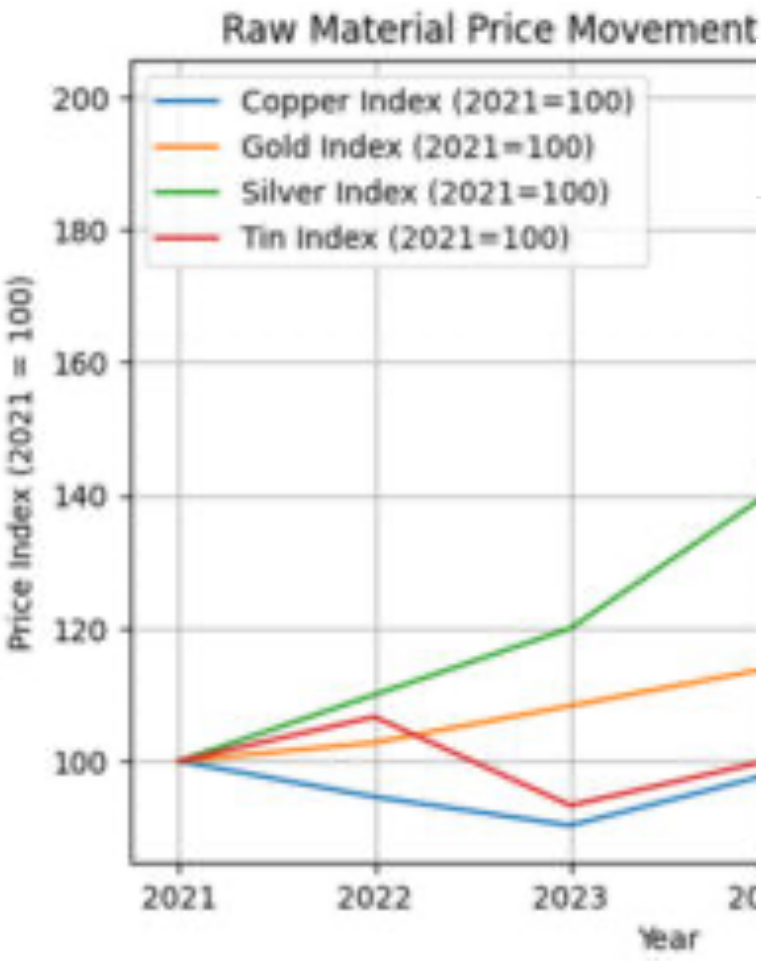
Medical Devices
Implantable electronics, sensor patches, minimally-invasive devices and portable diagnostic modules require ultra-thin, reliable interconnects for patient comfort and performance.

Communications & Mobile Electronics
Foldable smartphones, ultra-thin modules, high-density antennas and wear-on-body devices require flexible, ultra-thin interconnects to meet size, weight and performance goals.

Industrial, Aerospace & Defense
In mission-critical, weight-sensitive applications (drones, satellites, space wearables), ultra-thin flex circuits allow lighter assemblies and higher integration density.

Sensor & IoT Modules
Miniature sensors, body-worn systems, barcode scanners, smart cards and interconnected modules benefit from ultra-thin circuits with minimal mass and footprint.

The blogs featured below expand on this page's content, offering detailed insights into specific design, manufacturing, and application topics that provide added relevance and deeper context for engineers and decision-makers.



PCB, Flex & PCBA Cost Pressures to Watch

Electronics manufacturing is heading into 2026 under p g metals costs, laminate and prepreg repricing,...

April 7, 2026 By Nguyen Tran

Products/Technologies

Expertise

- Dynamic Flex
- Extra-long Flex Printed Circuits (FPC)
- Ultra-Thin Flexible Printed Circuits
- R2R (Reel to Reel Processing)
- Fineline PCB (High Density Interconnect)
- Blind and Buried Microvias
- Integrate Circuit Substrate Carrier
- Thermal Management





Integrated Flex Circuits and Injection Molding for Electronic Devices

At PICA Manufacturing Solutions, we believe the future of electronics is not built component by component, it is engineered as...

April 7, 2026 By Jim Lv



How Flex PCBs Are Shielded for EMI and RFI Protection

Electromagnetic interference (EMI) and radio frequency interference (RFI) can disrupt signal quality, create data errors, and reduce...

March 31, 2026 By FlexPros



How to Choose Flex PCB Materials for Bend Reliability, Heat, and Cost

A practical step-by-step method for choosing materials that hold up in bending, assembly, and real environmentsFlex material...

March 17, 2026 By Mona Ni

Next-Generation Wearable Health Monitoring: Polymer Thick Film Electronics and Biosensor Patches

Polymer Thick Film (Printed Electronics): Thin, Flexible, and Designed for the Human BodyPolymer Thick Film (PTF) technology, commonly referred to...

December 11, 2025 By Pascal Delloue

Ultra-Thin Flexible Circuits: Overcoming Challenges in Dielectric Substrate

In Part 1, we explored the challenges of manufacturing ultra-thin flexible circuits with a focus on the conductive layers, highlighting...

January 9, 2025 By FlexPros

Building Ultra-Thin Flexible Circuits: Part 1 – Conductor

One of our key customers approached PICA with the challenge of manufacturing ultra-thin flexible circuits. A critical component of this...

December 17, 2024 By FlexPros

Adhesiveless Technologies for Flexible Circuits: Unlocking Next-Generation Performance

In the world of flexible electronics, the shift from traditional flexible copper-clad laminates (FCCLs) to adhesiveless materials represents a significant...

December 9, 2024 By Simon Lim